

JEE Advanced Booster Test - 3 | 2022

Date: 07/12/2020

Maximum Marks: 180

Timing: 04:00 PM - 09:00 PM

Duration: 3.0 Hrs

General Instructions

- The question paper consists of 3 Parts (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **four** sections (**Section 1, Section 2, Section 3 and Section 4**).
- Section 1** contains **6 Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

Section 2 contains **2 Paragraphs containing 2 Questions** on each paragraph. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

Section 3 contains **2 Column-Match sets**. Each set contains statements given in two columns (I and II). One or more statements in column II can be matched with a statement in column I. Each set has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.

Section 4 contains **6 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code**, **Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

MARKING SCHEME

SECTION - 1

- This section consists of **SIX (06)** Questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+4 If only (all) the correct option(s) is(are) chosen
Partial Marks:	+3 If all the four options are correct but ONLY three options are chosen
Partial Marks:	+2 If three or more options are correct but ONLY two options are chosen and both of which are correct
Partial Marks:	+1 If two or more options are correct but ONLY one option is chosen, and it is a correct option
Zero Mark:	0 if none of the options is chosen (i.e. the question is unanswered)
Negative Marks:	-1 In all other cases.

SECTION - 2

- This section has **TWO (02) Paragraphs containing TWO (02) Questions** on each paragraph. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+3 If ONLY the correct option is chosen
Zero Marks:	0 If none of the options is chosen (i.e. the question is unanswered)
Negative Marks:	-1 In all other cases.

SECTION - 3

- This section has **TWO (02) Column-Match sets**. Each set contains statements given in two columns. Statements in column I have to be matched with statements in column II. One or more than one statements in column II can be matched with a statement in column I.
- Each set has **FOUR** options. **ONLY ONE** of these four options is the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+3 If ONLY the option corresponding to the correct combination is chosen
Zero Marks:	0 If none of the options is chosen (i.e., the question is unanswered)
Negative Marks:	-1 In all other cases.

SECTION 4

- This section contains **6 Integer Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks:	+3 If ONLY the correct Integer value is entered. There is NO negative marking .
Zero Marks:	0 In all other cases.

PART I : PHYSICS**60 MARKS****SECTION 1**

This section consists of SIX (06) Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

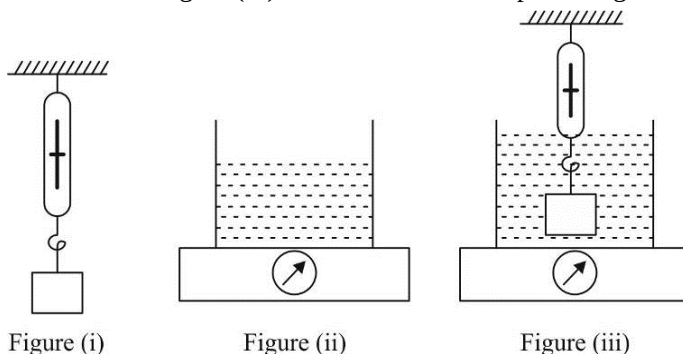
- A cyclist goes around a curve of radius r on a horizontal road with speed v . Which of the following options are correct?

(A) The required centripetal acceleration is provided by component of normal reaction

(B) The required centripetal acceleration is provided by friction

(C) The net force by the road on cyclist should pass through the centre of mass of the system of cycle and cyclist

(D) The angle made by the cyclist with the vertical while going around the curve is $\tan^{-1}\left(\frac{v^2}{rg}\right)$
- A spring balance gives a reading of 40 N when a block of relative density 2 is suspended from it as shown in figure (i). A weighing machine gives a reading of 100 N when a container having water is placed on it as shown in figure (ii). Now the block is submerged in water while still being suspended by the spring balance as shown in figure (iii). Choose the correct options regarding the final situation.



- (A) The reading of spring balance is now 20 N
- (B) The reading of weighing machine is now 140 N
- (C) The reading of weighing machine is now 120 N
- (D) If in the final situation, the block is removed from spring balance so that it finally settles on the bottom of container then reading of the weighing machine becomes 140 N
- In a free space, three identical particles of mass m each are initially held at the vertices of an equilateral triangle of side length l and then they are all released simultaneously. They start moving due to their mutual gravitational attraction. Choose the correct statements.

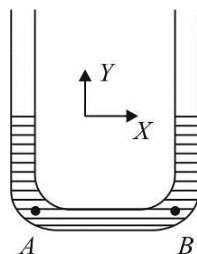
(A) The initial acceleration of a particle is $\frac{Gm}{l^2}$

(B) The initial acceleration of a particle is $\sqrt{3}\frac{Gm}{l^2}$

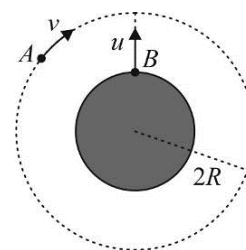
(C) The three particles always remain at the vertices of an equilateral triangle and the size of triangle keeps decreasing

(D) The speed of a particle after the separation between the particles reduces to half of the initial separation is $\sqrt{\frac{2Gm}{l}}$

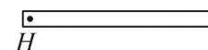
4. A U-tube of narrow cross section is kept in vertical xy plane and contains an incompressible liquid. A and B are points at the two ends of the horizontal portion of the tube as shown. Which of the following options are correct?



- (A) If the tube is accelerated vertically along $+y$ axis, pressure at A and B are equal
 (B) If the tube is accelerated horizontally along $+x$ axis, pressure at A is more than pressure at B
 (C) If the tube is accelerated horizontally along $+z$ axis, pressure at B is more than pressure at A
 (D) If the tube is rotated about a vertical axis parallel to y axis and passing through A , pressure at A is more than pressure at B
5. A satellite A of mass m is moving in a circular orbit of radius $2R$ around the earth. Another projectile B also of mass m is fired from the surface of earth with speed u in radial direction such that it strikes A and sticks to it. Choose the correct options.
 (M is the mass of earth, R is the radius of earth and G is the universal gravitational constant)



- (A) The orbital speed of satellite A is $\sqrt{\frac{GM}{2R}}$
 (B) The minimum speed of the combined mass after collision so that it escapes the gravity of earth is $\sqrt{\frac{GM}{R}}$
 (C) The minimum firing speed u of B so that combined mass is just able to escape the gravity of earth is $\sqrt{\frac{7}{2} \frac{GM}{R}}$
 (D) The minimum firing speed u of B so that combined mass is just able to escape the gravity of earth is $\sqrt{\frac{9}{2} \frac{GM}{R}}$
6. A thin uniform rod of mass m and length l is hinged at one end. It is released from horizontal position as shown in the figure. The hinge is smooth. Choose the correct options.



- (A) The initial angular acceleration of the rod is $\frac{3g}{2\ell}$
 (B) The force exerted by hinge on the rod just after release is $\frac{mg}{4}$
 (C) The angular velocity of the rod when it becomes vertical is $\sqrt{\frac{3g}{\ell}}$
 (D) The force exerted by hinge on the rod when it becomes vertical is mg

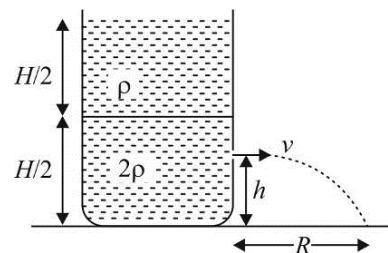
SPACE FOR ROUGH WORK

SECTION 2

This section has TWO (02) Paragraphs containing TWO (02) Questions on each paragraph. Each question has FOUR options. ONLY ONE of these four options is the correct answer.

Paragraph for Q.7 to Q.8

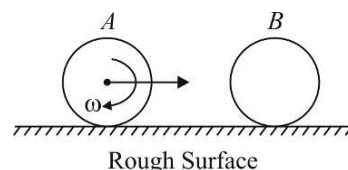
A container of large uniform cross-section area resting on a horizontal surface holds two immiscible, non-viscous and incompressible liquids of densities ρ and 2ρ . The height of both the liquid columns in the container is $H/2$ as shown in the figure. A small hole is punched at a height $h (< H/2)$ above the bottom.



7. Find the initial velocity of efflux of the liquid from the hole.
- (A) $\sqrt{2g(H-h)}$ (B) $\sqrt{2g(1.5H-2h)}$
 (C) $\sqrt{g(1.5H-2h)}$ (D) $\sqrt{g(2h-1.5h)}$
8. At what height h should the hole be punched so that the initial range R of the emerging stream is maximum?
- (A) $\frac{H}{4}$ (B) $\frac{H}{8}$ (C) $\frac{3H}{8}$ (D) $\frac{3H}{16}$

Paragraph for Q.9 to Q.10

A solid sphere A is rolling without slipping on a rough surface with angular speed ω as shown in figure. It collides elastically with another identical sphere B at rest. Assume no friction acts between the spheres during collision. Radius of the spheres is R .



9. Which of these statements is correct about the motion of the two spheres sometime after the collision?
- (A) A and B move in opposite directions
 (B) A and B move in same direction
 (C) A stops permanently and only B moves
 (D) A continues to rotate at a fixed location and B moves ahead without rotating
10. A long time after the collision, the ratio of velocity of centers of the spheres A and B is:
- (A) Zero (B) 1 (C) 0.4 (D) 0.67

SPACE FOR ROUGH WORK

SECTION 3

This section contains 2 Column-Match sets. Each set contains statements given in two columns (I and II). One or more statements in column II can be matched with a statement in column I. Each set has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

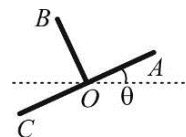
11. A satellite of mass m is in a circular orbit of radius a around a spherical planet of uniform density ρ and radius R . Match the columns.

COLUMN-I		COLUMN-II	
(I)	The time period of the satellite is proportional to	(P)	$\rho^{-1/2} R^{3/2} a^{3/2} m^0$
(II)	The force on the satellite is proportional to	(Q)	$\rho^1 R^3 a^{-2} m^1$
(III)	The kinetic energy of the satellite is proportional to	(R)	$\rho^{-1/2} R^{-3/2} a^{3/2} m^0$
(IV)	The total energy of the satellite is proportional to	(S)	$\rho^1 R^3 a^{-1} m^1$

Codes:

- (A) (I)-R; (II)-S; (III)-Q; (IV)-S
 (B) (I)-R; (II)-Q; (III)-S; (IV)-S
 (C) (I)-P; (II)-S; (III)-Q; (IV)-Q
 (D) (I)-P; (II)-Q; (III)-S; (IV)-S

12. Three thin uniform rods OA , OB and OC of mass $2m$, m and m respectively, and equal length, are joined at O to form a planar rigid body such that OB is perpendicular to both OA and OC . The rigid body is pivoted at the point O as shown, such that it can rotate about a horizontal axis passing through O . It is desired that the body remains in equilibrium in the position shown, with OA making an angle $\theta = \sin^{-1}\left(\frac{3}{5}\right)$ with the horizontal. This is achieved by the application of a force



F at a chosen point on the body.

Match the columns.

COLUMN-I		COLUMN-II	
(I)	If F is applied horizontally at A , its magnitude must be	(P)	$\frac{1}{2}mg$
(II)	If F is applied vertically at B , its magnitude must be	(Q)	$\frac{1}{5}mg$
(III)	If F is applied at the mid-point of OC , perpendicular to OC , its magnitude must be	(R)	$\frac{1}{6}mg$
(IV)	If F is applied vertically at C , its magnitude must be	(S)	$\frac{1}{8}mg$

Codes:

- (A) (I)-S; (II)-S; (III)-Q; (IV)-P
 (B) (I)-P; (II)-S; (III)-P; (IV)-R
 (C) (I)-R; (II)-R; (III)-Q; (IV)-S
 (D) (I)-R; (II)-R; (III)-Q; (IV)-P

SPACE FOR ROUGH WORK

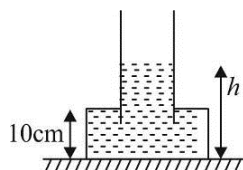
SECTION 4

This section has Six (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.*

(Example: 6, 81, 1.50, 3.25, 0.08)

13. Acceleration due to gravity is one-fourth of that on the surface of the earth at a height h above the surface and also at depth d below the surface. The ratio d/h is _____.

14. A circular hole of radius 10 cm is made in the base of a cylindrical container of radius 20 cm and height 10cm. A cylindrical pipe of radius 10 cm is inserted in the hole and the apparatus thus formed has a total mass of π kg. It is then kept inverted on a plain horizontal surface and water is poured into it. Assume that there is no leakage of water from bottom or the joint. Find the height h (in cm) of the water level when the apparatus gets lifted off the surface. (Density of water is 1000 Kg/m^3)



15. A uniform hollow cylinder rolls down a sufficiently rough inclined plane with acceleration a_H . A uniform solid cylinder rolls down the same incline with acceleration a_S . Find the ratio $\frac{a_H}{a_S}$.

16. A particle of mass m is placed at a distance l from one end of a uniform thin rod of mass M and length l as shown. The gravitational force between them is $\frac{GMm}{n\ell^2}$ where n is _____.



17. A large tank is filled with water to a height H . A small hole is made at the base of the tank. It takes T_1 time for the height of water to fall to $\frac{H}{9}$ and it takes further T_2 time for draining the rest of water. Then the ratio $\frac{T_1}{T_2}$ is _____.

18. The moment of inertia of a uniform hollow cubical box of mass M and side a about an axis passing through the centre of two opposite faces is equal to $x Ma^2$, where x is _____.

SPACE FOR ROUGH WORK

PART II : CHEMISTRY**60 MARKS****SECTION 1**

This section consists of SIX (06) Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

- One mole of a mono-atomic ideal gas is transformed from 300 K and 2 atm to 600 K and 4 atm. The entropy change for this process is :
 (A) $\frac{3}{2}R \ln 2$ (B) $\frac{1}{2}R \ln 2$ (C) $\frac{7}{2}R \ln 2$ (D) $\frac{5}{2}R \ln 2$
- The endothermic reaction shown below is at equilibrium in a sealed flask, with significant amounts of both $\text{Ca(OH)}_2(\text{s})$ and $\text{CaO}(\text{s})$ present. Which action will not increase the amount of $\text{Ca(OH)}_2(\text{s})$ at equilibrium ?

$$\text{Ca(OH)}_2(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{H}_2\text{O}(\text{g}) \quad \Delta H^\circ > 0$$
 (A) Crushing the $\text{Ca(OH)}_2(\text{s})$ into smaller pieces
 (B) Decreasing the temperature of the flask
 (C) Adding more $\text{CaO}(\text{s})$ to the flask
 (D) Adding $\text{N}_2(\text{g})$ to the flask
- Which reaction occurs with increase in entropy?
 (A) $\text{ClF}(\text{s}) \rightarrow \text{ClF}(\text{g})$
 (B) $\text{ClF}(\text{s}) \rightarrow \text{ClF}(\text{l})$
 (C) $\text{ClF}(\text{l}) \rightarrow \text{ClF}(\text{g})$
 (D) $\text{ClF}(\text{s}) \rightarrow \frac{1}{2}\text{Cl}_2(\text{g}) + \frac{1}{2}\text{F}_2(\text{g})$
- A sample of water in a movable piston is maintained at 100.0°C with an external pressure p . Which statement best describes the transformation of some of the liquid water into water vapor ?
 (A) ΔG is zero if $P = 1 \text{ atm}$
 (B) ΔS is positive for the transformation of some of the liquid water into water vapour
 (C) Strength of H-bond in the liquid water is more than in the water vapour
 (D) It is reversible if $p = 1 \text{ atm}$ and non-spontaneous if $p > 1 \text{ atm}$
- Which of the following reaction/process is exothermic ?
 (A) $\text{C}(\text{s}) (\text{diamond}) \longrightarrow \text{C}(\text{s}) (\text{graphite})$
 (B) $\text{P}(\text{white, solid}) \longrightarrow \text{P}(\text{Black, solid})$
 (C) Reversible process for a system ; which cause increase in entropy of the surrounding
 (D) Freezing of water at 0°C and 1 atm
- The reason why reactions come to an equilibrium point instead of going entirely to products is :
 (A) The attractive and repulsive forces between molecules balance exactly at that point
 (B) The reaction is always endothermic at that point
 (C) The reaction is always exothermic at that point
 (D) The free energy of the reaction mixture is minimum at this point

SECTION 2

options. ONLY ONE of these four options is the correct answer.

Paragraph for O-7 to O-8

Spontaneous - is a reaction that is considered to be natural because it is a reaction that occurs by itself without any external action towards it. Non-spontaneous needs constant external energy applied to it in order for the process to continue and once you stop the external action the process will cease. When solving for the equation, if change of G is negative, then it's spontaneous. If change of G is positive, then it's non-spontaneous. The symbol that is commonly used for FREE ENERGY is G.

$$\Delta G \text{ (change in free energy)} = \Delta H \text{ (change in enthalpy)} - T\Delta S \text{ (temperature) (change in entropy)}$$

- I. ΔG depends only on the difference in free energy of products and reactants (or final state and initial state). ΔG is independent of the path of the transformation and is unaffected by the mechanism of a reaction.
- II. ΔG cannot tell us anything about the rate of a reaction.

The standard-state free energy of reaction (ΔG°) is defined as the free energy of reaction at standard state conditions :

$$\Delta G^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ}$$

7. For a reaction studied over a wide temperature range, plots of $\frac{\Delta G_{\text{Reaction}}^{\circ}}{T}$ vs $\frac{1}{T}$ and of $\Delta G_{\text{Reaction}}^{\circ}$ vs T are linear. Which of the following must be independent of temperature?

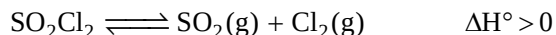
- | | | | |
|------------|--------------------------------------|------------|--------------------------------------|
| I. | $\Delta G^{\circ}_{\text{Reaction}}$ | II. | $\Delta H^{\circ}_{\text{Reaction}}$ |
| (A) | I only | (B) | II only |
| (C) | Both I and II | (D) | Neither I nor II |

8. Elemental tin forms two allotropes, white tin and gray tin. For the conversion of white tin to gray tin, $\Delta H^\circ = -2.02 \text{ KJ mol}^{-1}$ and $\Delta S^\circ = -7.04 \text{ J mol}^{-1} \text{ K}^{-1}$. Which statement best describes the stability of the two phases ? [Given that $273.15 = 0^\circ\text{C}$]
- (A) White tin is stable below 13.8°C and gray tin is stable above this temperature
- (B) Gray tin is stable below 13.8°C and white tin is stable above this temperature
- (C) White tin is stable below 0.3°C and gray tin is stable above this temperature
- (D) Gray tin is stable below 0.3°C and white tin is stable above this temperature

Paragraph for Q-9 to Q-10

Le Chatelier's principle states that if a dynamic equilibrium is disturbed by changing the conditions, the position of equilibrium shifts to counteract the change to re-establish an equilibrium. If a chemical reaction is at equilibrium and experiences a change in pressure, temperature, or concentration of products or reactants, the equilibrium shifts in the opposite direction to offset the change.

9. A sample of SO_2Cl_2 is introduced into a movable piston with the pressure maintained at 1.00 atm and the temperature at 450 K. The volume increases as the sulfuryl chloride equilibrates according to the following endothermic reaction:



Which changes would decrease the number of moles of $\text{SO}_2\text{Cl}_2(\text{g})$ present in the piston at equilibrium ?

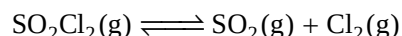
In each case, the pressure is maintained at 1.00 atm.

I. The temperature of the system is increased to 500 K.

II. A sample of $\text{Ar}(\text{g})$ is injected into the piston.

(A) I only (B) II only (C) Both I and II (D) Neither I nor II

10. Sulfuryl chloride is in equilibrium with sulfur dioxide and chlorine gas :



A system with a volume of 1.00 L is in equilibrium at a certain temperature with $p(\text{SO}_2\text{Cl}_2) = 1.00$ bar and $p(\text{SO}_2) = p(\text{Cl}_2) = 0.10$ bar. By how much will the number of moles of SO_2Cl_2 at equilibrium change if the volume is reduced to 0.50 L?

(A) Increase 1-10% (B) Increase 11-50%
(C) Decrease 1-10% (D) Decrease 11-50%

SPACE FOR ROUGH WORK

SECTION 3

This section contains 2 Column-Match sets. Each set contains statements given in two columns (I and II). One or more statements in column II can be matched with a statement in column I. Each set has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

11. MATCH THE FOLLOWING:

	Reaction		Change in Thermodynamics Function
I.	$3\text{O}_2(\text{g}) \longrightarrow 2\text{O}_3(\text{g})$	(p)	$\Delta_r G < 0$ (always)
II.	$2\text{C}(\text{Solid, graphite}) + \text{O}_2(\text{g}) \longrightarrow 2\text{CO}(\text{g})$	(q)	$\Delta S_{\text{uni}} < 0$ (always)
III.	$\text{P}(\text{Solid, white}) \longrightarrow \text{P}(\text{Solid, Red})$	(r)	$\Delta_r H < 0$ and $\Delta_r S < 0$
IV.	$\text{NH}_4\text{Cl}(\text{s}) \longrightarrow \text{NH}_3(\text{g}) + \text{HCl}(\text{g})$	(s)	$\Delta_r H > 0$ and $\Delta_r S > 0$

	I	II	III	IV		I	II	III	IV
(A)	p, r	p, s	q, r	r, s	(B)	q	p	r	s
(C)	q, s	p, r	p, r	p, s	(D)	q, r, s	p, r, s	p, q	q, r, s

12. MATCH THE FOLLOWING

	Process		Variable
I.	Isothermal expansion of an ideal gas	(p)	$w = -q$
II.	Free expansion	(q)	$w = 0$
III.	Adiabatic expansion	(r)	$\Delta U = 0$
IV.	Isochoric	(s)	$q = 0$

	I	II	III	IV		I	II	III	IV
(A)	p, r	q, r, s	s	q	(B)	p, s	s	r, s	q
(C)	q, s	p, s	q, r	q	(D)	q, s	p, s	q, r	q

SPACE FOR ROUGH WORK

SECTION 4

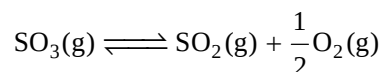
This section has Six (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.*

(Example: 6, 81, 1.50, 3.25, 0.08)

13. At equilibrium, a solution contains 1.0 mol each of $\text{Cd}^{2+}(\text{aq})$, $\text{SCN}^{-}(\text{aq})$ and $\text{Cd}(\text{SCN})_4^{2-}(\text{aq})$. What is the solution's volume ?



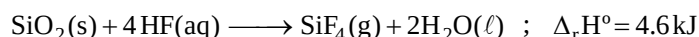
14. At 400 K, this reaction has $K_p = 8.2 \times 10^{-4}$.



K_p at 400 K for the following reaction.

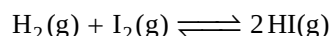


15. The standard enthalpy of reaction for the dissolution of silica in aqueous HF is 4.6 kJ mol^{-1} . What is the standard enthalpy of formation (in kJ) of $\text{SiF}_4(\text{g})$?



Compound	$\text{SiO}_2(\text{s})$	$\text{HF}(\text{aq})$	$\text{H}_2\text{O}(\ell)$	$\text{SiF}_4(\text{g})$
$\Delta_f H^\circ, \text{ kJ mol}^{-1}$	-910.9	-320.1	-285.8	???

16. The following reaction has $K_p = 50.41$ at 448°C .



If a 3.00 L flask initially contains 0.0500 moles each of H_2 and I_2 , how many moles of HI are present when the contents have reached equilibrium at 448°C ?

17. 40.0 mL of 0.200 M aqueous NaOH is added to 200.0 mL of 0.100 M aqueous NaHCO_3 in a flask maintained at 25°C . Neglecting the effects of dilution, what is $q(\text{in J})$ for this reaction ?

	$\Delta_f H^\circ, \text{ kJ mol}^{-1}$
$\text{OH}^{-}(\text{aq})$	-230
$\text{HCO}_3^{-}(\text{aq})$	-692
$\text{CO}_3^{2-}(\text{aq})$	-677
$\text{H}_2\text{O}(\ell)$	-286

18. In a well-insulated vessel, 50.0 g ice at 0.0°C is added to 350 g water at 32.0°C . What is the final temperature in degree celsius? (The heat of fusion of ice is 334.72 J g^{-1} and specific heat capacity of water is $4.184 \text{ J / g}^\circ\text{C}$)

PART III : MATHEMATICS**60 MARKS****SECTION 1**

This section consists of SIX (06) Questions. Each question has FOUR options. ONE OR MORE THAN ONE of these four option(s) is(are) correct answer(s).

- The coordinates of A, B, C are (6, 3), (−3, 5) and (4, −2) respectively, then:
 - Origin lies outside $\triangle ABC$
 - Origin lies inside $\triangle ABC$
 - Orthocentre of $\triangle ABC$ does not lie inside $\triangle ABC$
 - Orthocentre lies on $3x + 4y = 22$
- Two sides of a rhombus ABCD are parallel to the lines $y = x + 2$ and $y = 7x + 3$. If diagonals of the rhombus intersect at the point (1, 2) & vertex A is on y-axis, then possible coordinates of B can be:
 - $\left(\frac{1}{3}, \frac{7}{3}\right)$
 - $\left(\frac{5}{3}, \frac{5}{3}\right)$
 - $\left(-\frac{1}{2}, -1\right)$
 - $\left(\frac{5}{2}, 5\right)$
- Consider two circles $x^2 + y^2 - 4x - 2y + 4 = 0$ and $x^2 + y^2 - 12x - 8y + 36 = 0$, then which of the following is/are true?
 - two circles touch each other at (a, b) such that $2a - b = 4$
 - one of the common tangents is x-axis
 - one of the common tangents is y-axis
 - area of Δ formed by the common tangents is $\frac{16}{3}$ sq. units
- A circle touching the line $4x - y = 4$ at (2, 4) & passing through the point (0, 1) satisfies:
 - Centre of the circle is $\left(-\frac{16}{5}, \frac{53}{10}\right)$
 - Centre of the circle is $\left(-\frac{16}{5}, -\frac{27}{10}\right)$
 - angle made by chord joining (2, 4) & (0, 1) at centre is $2 \tan^{-1}\left(\frac{5}{14}\right)$
 - angle made by chord joining (2, 4) & (0, 1) at centre is $2 \tan^{-1}\left(\frac{11}{10}\right)$
- For $r = 0, 1, 2, \dots, 20$, let A_r, B_r, C_r denote respectively the coefficient of x^r in the expansions of $(1+x)^{20}$, $(1+x)^{40}$ and $(1+x)^{60}$, then:
 - $\sum_{r=1}^{20} A_r (B_r B_{20} - C_{20} A_r) = C_{20} - B_{20}$
 - $\sum_{r=0}^{20} A_r (B_r B_{20} - C_{20} A_r) = 0$
 - $\sum_{r=0}^{20} A_r C_r = {}^{80}C_{20}$
 - $\sum_{r=0}^{20} (A_r + B_r) - {}^{39}C_{20} = 2^{20} (2^{19} + 1)$
- Consider the expansion of $(1 - 2\sqrt{x})^{50}$, then sum of the coefficients of integral powers of x is:
 - $\frac{3^{50} + 1}{2}$
 - divisible by 2
 - $\frac{3^{50} - 1}{2}$
 - divisible by 5

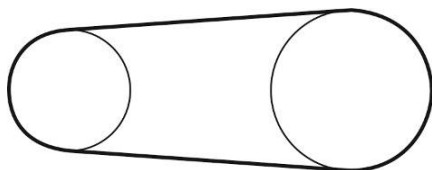
SPACE FOR ROUGH WORK

SECTION 2

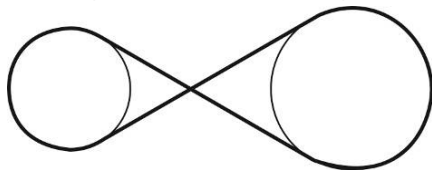
This section has TWO (02) Paragraphs containing TWO (02) Questions on each paragraph. Each question has FOUR options. ONLY ONE of these four options is the correct answer.

Paragraph for Q-7 to Q-8

Consider 2 circular wheels $x^2 + y^2 = 1$ & $(x-6)^2 + y^2 = 4$, there is a belt 1, along direct common tangents as shown below



and belt 2, along indirect common tangents as shown below



7. Length of belt 1 is given by:

- (A) $2(\sqrt{35} + \pi - \tan^{-1} \sqrt{35})$ (B) $2(\sqrt{35} + 2\pi - \tan^{-1} \sqrt{35})$
 (C) $2(\sqrt{35} + 3\tan^{-1} \sqrt{35})$ (D) None of these

8. Length of belt 2 is given by:

- (A) $6\sqrt{3} + 4\pi$ (B) $6\sqrt{3} + \frac{8\pi}{3}$ (C) $6\sqrt{3} + \frac{4\pi}{3}$ (D) None of these

Paragraph for Q-9 to Q-10

To find sum of series involving binomial coefficients, we consider some binomial expansion involving these coefficients, manipulate them & put suitable values of variables involved.

For Example, to find or simplify ${}^{19}C_1 2^2 + {}^{19}C_3 2^6 + {}^{19}C_5 2^{10} + \dots + {}^{19}C_{19} 2^{38}$

We consider $(1+x)^{19} = {}^{19}C_0 + {}^{19}C_1 x + \dots + {}^{19}C_{19} x^{19}$ & $(1-x)^{19} = {}^{19}C_0 - {}^{19}C_1 x + \dots - {}^{19}C_{19} x^{19}$

Subtract $(1+x)^{19} - (1-x)^{19} = 2({}^{19}C_1 x + {}^{19}C_3 x^3 + \dots + {}^{19}C_{19} x^{19})$

Let $x = 4$

$$\Rightarrow {}^{19}C_1 4 + {}^{19}C_3 4^3 + \dots + {}^{19}C_{19} 4^{19} = \frac{5^{19} - 3^{19}}{2}$$

9. $(1 \cdot {}^{20}C_1)^2 - (2 \cdot {}^{20}C_2)^2 + (3 \cdot {}^{20}C_3)^2 - \dots + (19 \cdot {}^{20}C_{19})^2 - (20 \cdot {}^{20}C_{20})^2$

- (A) ${}^{20}C_{10}$ (B) $\frac{1}{2} {}^{20}C_{10}$ (C) 0 (D) $11 \cdot {}^{20}C_{10}$

10. $\sum_{r=0}^{10} (-1)^r \frac{{}^{10}C_r}{{}^{r+3}C_r} =$

- (A) $\frac{3}{13}$ (B) $\frac{6}{13}$ (C) $\frac{3}{26}$ (D) None of these

SPACE FOR ROUGH WORK

SECTION 3

This section contains 2 Column-Match sets. Each set contains statements given in two columns (I and II). One or more statements in column II can be matched with a statement in column I. Each set has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

11. MATCH THE FOLLOWING COLUMNS:

	Column I		Column II
I.	The lines $y = 0, y = 1, x - 6y + 4 = 0$ & $x + 6y - 9 = 0$ constitute a figure which is:	(p)	A cyclic quadrilateral
II.	The points $A(a, 0), B(0, b), C(c, 0)$ & $D(0, d)$ are such that $ac = bd$ & $abcd \neq 0$. The points A, B, C & D always constitute	(q)	A rhombus
III.	The figure formed by the four lines $ax \pm by \pm c = 0 (a \neq b)$ is	(r)	A square
IV.	The lines $x = 2, x = 6, y = 5, y = 9$ constitute a figure which is	(s)	A trapezium

	I	II	III	IV		I	II	III	IV
(A)	p	s	s	p, s	(B)	s	p, s	s	p, q, r, s
(C)	p, s	p	q, s	p, q, r, s	(D)	p, s	p, s	q, s	q, s

12. MATCH THE FOLLOWING COLUMNS:

	Column I		Column II
I.	Number of irrational terms in the expansion of $(\sqrt{2} + \sqrt[3]{3})^{15}$ is $2k + 1$, then k is greater than:	(p)	5
II.	If $\sum_{r=1}^n {}^nC_r 3^r = 4095$, then n is less than	(q)	6
III.	If ${}^nP_r = ({}^{n-1}C_{r-1} + {}^{n-1}C_r) \cdot 5040$, then r is less than	(r)	7
IV.	If $\sum_{r=1}^{10} r(r-1) {}^{10}C_r = k \cdot 9 \cdot 2^9$, then k is less than	(s)	8

	I	II	III	IV		I	II	III	IV
(A)	p, q	r, s	r, s	p, q, r, s	(B)	p	q, r, s	s	r, s
(C)	p	r, s	s	q, r, s	(D)	p, q	q, r, s	r, s	p, q, r, s

SPACE FOR ROUGH WORK

SECTION 4

This section has Six (06) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.*

(Example: 6, 81, 1.50, 3.25, 0.08)

13. Coefficient of x^4 in the expansion of $\left(\frac{x}{2} - \frac{3}{x^2}\right)^{10}$ is _____.
14. The number of terms in the expansion of $(1+x)^{101}(1+x^2-x)^{100}$ in powers of x is _____.
15. If λ is integer and $P(\lambda, 2)$ lies inside $\triangle ABC$, formed by the lines $x+y=4$, $3x-7y=8$ and $4x-y=31$, then λ is _____.
16. If $P(1, 2)$, $Q(4, 6)$, $R(5, 7)$, $S(a, b)$ are vertices of a parallelogram $PQRS$, then $\sqrt{a+b}$ is _____.
17. Tangent are drawn from the point $P(1, 8)$ to the circle $x^2 + y^2 - 6x - 4y - 11 = 0$, touching the circle at the points A & B, then area of $\triangle PAB$ is $k\sqrt{6}$ sq. units, then k is _____.
18. The centre of the circle passing through the points $(0, 0)$, $(1, 0)$ & touching the circle $x^2 + y^2 = 16$ is $\left(\frac{1}{2}, k\right)$, then k^2 is _____.

SPACE FOR ROUGH WORK

🌀 🌀 🌀 End of JEE Advanced Booster Test - 3 | 2022 🌀 🌀 🌀